

1 Gain law

(bars denote normalized quantities: lengths by R ; $a_o := \Delta t / (\gamma_N R)$ [1/pN] and $a_{\text{nom}} := \Delta t / \gamma_N = a_o R$ [$\mu\text{m}/\text{pN}$]; $\bar{a} := a/a_{\text{nom}}$; forces by a_o : $\bar{f} := a_o f$, so that $\bar{a} \bar{f} = \Delta \bar{w}$; $0 \leq \bar{a} \leq 1$, far field 1, contact 0; hats sit on the normalized quantities $\hat{a}_w, \hat{b}, \delta \hat{w}$; $e_\theta = \theta - \hat{\theta}$)

$$\begin{aligned}\bar{a}_{\text{true}}(\bar{w}) &= 1 - \frac{9}{8\bar{w}} + \frac{1}{2\bar{w}^3} - \frac{57}{100\bar{w}^4} + \frac{1}{5\bar{w}^5} + \frac{7}{200\bar{w}^{11}} - \frac{1}{25\bar{w}^{12}} \\ \bar{a}'_{\text{true}}(\bar{w}) &= \frac{d\bar{a}_{\text{true}}}{d\bar{w}} \\ \bar{a}''_{\text{true}}(\bar{w}) &= \frac{d^2\bar{a}_{\text{true}}}{d\bar{w}^2}\end{aligned}$$

$$\bar{a}'_{\text{true}}(\bar{w}) > 0 \text{ on } \bar{w} > 1 \implies \bar{a}' = b_{\text{true}}(\bar{a})(1 - \bar{a})^2 \text{ exactly,} \quad b_{\text{true}} := \frac{\bar{a}'_{\text{true}}}{(1 - \bar{a}_{\text{true}})^2}$$

$$\bar{a}_{\text{true}}(1) = 0, \bar{a}'_{\text{true}}(1) = 1 \implies b_{\text{true}}(\bar{a} = 0) = 1$$

$$1 - \bar{a}_{\text{true}} = \frac{9}{8\bar{w}} + \mathcal{O}(\bar{w}^{-3}) \implies b_{\text{true}}(\bar{a} \rightarrow 1) = \frac{8}{9}$$

$$\bar{a}''_{\text{true}}(\bar{w}) = b'_{\text{true}}(\bar{w})(1 - \bar{a}_{\text{true}})^2 - 2b_{\text{true}}(1 - \bar{a}_{\text{true}})\bar{a}'_{\text{true}}$$

$$\bar{a}(\bar{w}) = 1 - \frac{1}{b(\bar{w} - \bar{w}_s)}$$

$$\bar{a}'(\bar{a}, b) := b(1 - \bar{a})^2$$

$$\bar{a}''(\bar{a}, b) := \frac{d\bar{a}'}{d\bar{w}} = \frac{\partial \bar{a}'}{\partial \bar{a}} \bar{a}' = -2b(1 - \bar{a})\bar{a}' = -2b^2(1 - \bar{a})^3$$

$$a_w = a_o \bar{a}_w, \quad a_o = \frac{\Delta t}{\gamma_N R}$$

2 Jacobians

$$\begin{aligned}A := \frac{\partial \bar{a}'}{\partial \bar{a}} &= -2b(1 - \bar{a}) \\ \frac{\partial \bar{a}'}{\partial b} &= (1 - \bar{a})^2 = \frac{\bar{a}'}{b} \\ \frac{\partial \bar{a}''}{\partial \bar{a}} &= 6b^2(1 - \bar{a})^2 \\ \frac{\partial \bar{a}''}{\partial b} &= -4b(1 - \bar{a})^3 = \frac{2\bar{a}''}{b}\end{aligned}$$

3 True dynamics

$$(\delta \bar{w}_j[k] := \delta \bar{w}[k - d + j - 1], \quad j = 1, \dots, d + 1, \quad d = 2)$$

$$\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta \bar{w}_3[k] \\
\bar{w}_d[k+1] &= \bar{w}_d[k] + \Delta \bar{w}_d[k] \\
\delta \bar{w}_1[k+1] &= \delta \bar{w}_2[k] \\
\delta \bar{w}_2[k+1] &= \delta \bar{w}_3[k] \\
\delta \bar{w}_3[k+1] &= \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\Delta \bar{w}[k] &= \Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
\bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'_{\text{true}}[k] \Delta \bar{w}[k] + \frac{1}{2} \bar{a}''_{\text{true}}[k] \Delta \bar{w}[k]^2 \\
b_{\text{true}}[k+1] &= b_{\text{true}}[k] + b'_{\text{true}}[k] \Delta \bar{w}[k]
\end{aligned}$$

$$\begin{aligned}
F_{dw}[k] &= \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k-i] \\
\varepsilon_w[k] &= \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k-i] \bar{f}_T[k-i] - (1 - \lambda_c) n_w[k]
\end{aligned}$$

4 Process model

$$\mathbf{x} = [\delta \bar{w}_1, \delta \bar{w}_2, \delta \bar{w}_3, \bar{a}_w, b]^\top \quad (d=2)$$

5 Estimator

$$\begin{aligned}
\delta \hat{w}_1[k+1] &= \delta \hat{w}_2[k] \\
\delta \hat{w}_2[k+1] &= \delta \hat{w}_3[k] \\
\delta \hat{w}_3[k+1] &= \lambda_c \delta \hat{w}_3[k] \\
\Delta \hat{w}[k] &= \Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k] \\
\hat{a}_w[k+1] &= \hat{a}_w[k] + \hat{b}[k] (1 - \hat{a}_w[k])^2 \Delta \hat{w}[k] - \hat{b}[k]^2 (1 - \hat{a}_w[k])^3 \Delta \hat{w}[k]^2 + m_2[k] \\
\hat{b}[k+1] &= \hat{b}[k]
\end{aligned}$$

$$\frac{1}{1 - \hat{a}_w[k+1]} = \frac{1}{1 - \hat{a}_w[k]} + \hat{b}[k] \Delta \hat{w}[k] \quad (\text{exact step; equal to the two law terms above to second order})$$

6 Error dynamics

(True – Estimator; coefficients at the estimate; $\bar{a}' := \hat{b}(1 - \hat{a}_w)^2$, $A := -2\hat{b}(1 - \hat{a}_w)$, $\bar{a}'' := A \bar{a}'$;
 $e_b[k] := b_{\text{true}}[k] - \hat{b}[k]$, $f[k] := (1 - \hat{a}_w[k])^2 e_b[k]$, $f' := df/d\bar{w}$)

$$\begin{aligned}
e_{\Delta\bar{w}}[k] &= \Delta\bar{w}[k] - \Delta\hat{w}[k] = (1 - \lambda_c) e_{\delta\bar{w}_3}[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
\bar{a}'_{\text{true}}[k] &= \bar{a}' + A e_{\bar{a}_w}[k] + f[k] + \hat{b} e_{\bar{a}_w}[k]^2 - 2(1 - \hat{a}_w) e_{\bar{a}_w}[k] e_b[k] + e_b[k] e_{\bar{a}_w}[k]^2 \\
\bar{a}''_{\text{true}}[k] &= \bar{a}'' + f'[k] + \frac{\bar{a}''}{\bar{a}'} f[k] + \mathcal{O}(e)
\end{aligned}$$

$$\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c e_{\delta\bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= e_{\bar{a}_w}[k] + \underbrace{\bar{a}' e_{\Delta\bar{w}} + A \Delta\hat{w} e_{\bar{a}_w} + \Delta\hat{w} f}_{(1)} + \underbrace{A e_{\bar{a}_w} e_{\Delta\bar{w}} + f e_{\Delta\bar{w}} + \bar{a}'' \Delta\hat{w} e_{\Delta\bar{w}} + \frac{1}{2} \bar{a}'' e_{\Delta\bar{w}}^2}_{(2)} \\
&\quad + \underbrace{\frac{1}{2} \left(f' + \frac{\bar{a}''}{\bar{a}'} f \right) (\Delta\hat{w} + e_{\Delta\bar{w}})^2 + \hat{b} e_{\bar{a}_w}^2 (\Delta\hat{w} + e_{\Delta\bar{w}}) - 2(1 - \hat{a}_w) e_{\bar{a}_w} e_b (\Delta\hat{w} + e_{\Delta\bar{w}})}_{(3)} \\
&\quad - m_2 + e_b e_{\bar{a}_w}^2 (\Delta\hat{w} + e_{\Delta\bar{w}}) \\
e_b[k+1] &= e_b[k] + b'_{\text{true}}[k] \Delta\bar{w}[k]
\end{aligned}$$

((1) degree one without curvature; (2), (3) every term carrying $\bar{a}''$ or f' ; last term third order)

$$\begin{aligned}
\text{Var}(e_{\Delta\bar{w}}) &= (1 - \lambda_c)^2 P_{33} + F_{dw}^2 P_{44} + 2(1 - \lambda_c) F_{dw} P_{34} + Q_{33} \\
\text{Cov}(e_{\bar{a}_w}, e_{\Delta\bar{w}}) &= (1 - \lambda_c) P_{34} + F_{dw} P_{44} \\
\text{Cov}(e_b, e_{\Delta\bar{w}}) &= (1 - \lambda_c) P_{35} + F_{dw} P_{45} \\
m_2[k] &= A \text{Cov}(e_{\bar{a}_w}, e_{\Delta\bar{w}}) + \frac{1}{2} \bar{a}'' \text{Var}(e_{\Delta\bar{w}}) + \frac{\bar{a}' + \bar{a}'' \Delta\hat{w}}{\hat{b}} \text{Cov}(e_b, e_{\Delta\bar{w}}) \\
&\quad + \frac{1}{2} f' (\Delta\hat{w}^2 + \text{Var}(e_{\Delta\bar{w}})) + \hat{b} P_{44} \Delta\hat{w} - 2(1 - \hat{a}_w) P_{45} \Delta\hat{w}
\end{aligned}$$

First order, (1) with $e_{\Delta\bar{w}}$ substituted:

$$\begin{aligned}
e_{\bar{a}_w}[k+1] &= \left[1 + \hat{b}(1 - \hat{a}_w)^2 F_{dw} - 2\hat{b}(1 - \hat{a}_w) (\Delta\bar{w}_d + (1 - \lambda_c) \delta\hat{w}_3) \right] e_{\bar{a}_w}[k] \\
&\quad + (1 - \lambda_c) \hat{b}(1 - \hat{a}_w)^2 e_{\delta\bar{w}_3}[k] + (1 - \hat{a}_w)^2 (\Delta\bar{w}_d + (1 - \lambda_c) \delta\hat{w}_3) e_b[k] + \hat{b}(1 - \hat{a}_w)^2 \varepsilon_w[k] \\
e_b[k+1] &= e_b[k] + b'_{\text{true}}[k] \Delta\bar{w}[k]
\end{aligned}$$

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 \\ 0 & 0 & (1 - \lambda_c) \hat{b}(1 - \hat{a}_w)^2 & 1 + \hat{b}(1 - \hat{a}_w)^2 F_{dw} & (1 - \hat{a}_w)^2 \\ 0 & 0 & 0 & -2\hat{b}(1 - \hat{a}_w) (\Delta\bar{w}_d + (1 - \lambda_c) \delta\hat{w}_3) & \times (\Delta\bar{w}_d + (1 - \lambda_c) \delta\hat{w}_3) \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{q}[k] = \left[0, 0, -\varepsilon_w, \hat{b}(1 - \hat{a}_w)^2 \varepsilon_w, 0 \right]^\top$$

model error : row 4: $\Delta\hat{w} f = F_e(4, 5) e_b$ with $\mathbb{E}[e_b] \neq 0$; row 5: $b'_{\text{true}} \Delta\bar{w}$

7 Measurement and innovation

$$\nabla_d \bar{w}_d[k] = \bar{w}_d[k] - \bar{w}_d[k-d]$$

$$y_1[k] = \delta \bar{w}_1[k] + n_w[k]$$

$$y_2[k] = \bar{a}_w[k-d] + n_a[k]$$

$$\bar{a}_w[k-d] = \bar{a}_w[k] - b(1 - \bar{a}_w[k])^2 \nabla_d \bar{w}_d[k]$$

$$\frac{\partial y_2}{\partial \bar{a}_w} = 1 + 2\hat{b}(1 - \hat{a}_w) \nabla_d \bar{w}_d$$

$$\frac{\partial y_2}{\partial b} = -(1 - \hat{a}_w)^2 \nabla_d \bar{w}_d$$

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 + 2\hat{b}(1 - \hat{a}_w) \nabla_d \bar{w}_d & -(1 - \hat{a}_w)^2 \nabla_d \bar{w}_d \end{bmatrix}$$

$$L[k] = P^f H^\top (H P^f H^\top + R)^{-1}$$

$$e_{y_1}[k] = e_{\delta \bar{w}_1}[k] + n_w[k]$$

$$e_{y_2}[k] = \left[1 + 2\hat{b}(1 - \hat{a}_w) \nabla_d \bar{w}_d \right] e_{\bar{a}_w}[k] - (1 - \hat{a}_w)^2 \nabla_d \bar{w}_d e_b[k] + n_a[k]$$

8 Process and measurement noise

$$\kappa_T := \frac{4k_B T a_o}{R} = \frac{4k_B T a_{\text{nom}}}{R^2}$$

$$\text{Var}(\bar{a}_w[k] \bar{f}_T[k]) = \kappa_T \bar{a}_w[k] \rightarrow \kappa_T \hat{a}_w[k]$$

$$\sigma_{n_w}^2 = \frac{\sigma_n^2}{R^2}$$

$$Q_{33} = \text{Var}(\varepsilon_w)[k] = \kappa_T \left(\hat{a}_w[k] + (1 - \lambda_c)^2 (\hat{a}_w[k-1] + \hat{a}_w[k-2]) \right) + (1 - \lambda_c)^2 \sigma_{n_w}^2$$

$$Q_{34} = Q_{43} = -\hat{b}(1 - \hat{a}_w)^2 Q_{33}$$

$$Q_{44} = \hat{b}^2 (1 - \hat{a}_w)^4 Q_{33}$$

$$Q_{55} = 0$$

$$R = \text{diag}(R_1, R_2)$$

$$R_1 = \sigma_{n_w}^2$$

$$R_2 = K_{\text{var}} I F_{\text{var}} (\hat{a}_w + \xi)^2 + d Q_{44}$$

$$K_{\text{var}} = \frac{2a_{\text{cov}}}{2 - a_{\text{cov}}}$$

Seed and prior

$$\begin{aligned}\hat{b}[0] &= \frac{8}{9}, & P_{55}[0] &= \left(\sup_{\text{env}} |b_{\text{true}} - \frac{8}{9}| \right)^2 \\ \hat{a}_w[0] &= 1 - \frac{1}{\hat{b}[0](\bar{w}[0] - \bar{w}_s)}, & P_{44}[0] &= \bar{a}'[0]^2 P_{f,w_0}^2 + P_{f,a,\text{floor}}^2\end{aligned}$$